

Recap

Two more derivations

De Morgan's Law can be extended:

$$\neg(p_1 \vee p_2 \vee p_3 \vee p_4) \equiv \neg((p_1 \vee p_2) \vee (p_3 \vee p_4))$$

The above is a negation of a disjunction of two things.

$$\equiv \neg((p_1 \vee p_2) \vee p_3) \wedge \neg p_4$$

$$\equiv \neg((p_2 \vee p_2) \wedge \neg p_3) \wedge \neg p_4$$

De Morgan's applied ...

$$\equiv ((\neg p_1 \wedge \neg p_2) \wedge \neg p_3) \wedge \neg p_4$$

$$\equiv \neg p_1 \wedge \neg p_2 \wedge \neg p_3 \wedge \neg p_4$$

Parentheses are optional

Similarly

Wait for Chapter 5 so we can use induction (the “...” symbol)

$$\neg(p_1 \wedge p_2 \wedge p_3 \dots \wedge p_n) \equiv \neg p_1 \vee \neg p_2 \dots \vee \neg p_n$$

Example

Claim:

$$(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge (p_3 \rightarrow q) \equiv (p_1 \vee p_2 \vee p_3) \rightarrow q$$

Verification

$$(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge (p_3 \rightarrow q) \equiv ((p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge (p_3 \rightarrow q))$$

$$\equiv ((p_1 \vee p_2) \rightarrow q) \wedge (p_3 \rightarrow q)$$

^ By 7th rule in table 7 (from book) ^

$$\equiv ((p_1 \vee p_2) \vee p_3) \rightarrow q$$

Same rule on the whole expression.

Satisfiability

Definition:

- A expression is satisfiable iff, there is an assignment of T and F to its variables so it evaluates to T

This is exactly the opposite of being a contradiction,

1.4 Predicates and Quantifiers

$$x > 5$$

The above is not a proposition... Until we assign a value to x .

“blue > 5” still isn't a proposition

We need to pick a domain, a collection of possible values that will make the expression a proposition.

we use function notation:

Example

if

$$Q(x)$$

is

$$x + 7 = 13$$

then

$$Q(6)$$

is

$$6 + 7 = 13$$

which is true,

$$Q(5)$$

is not true

Propositional function

$$Q(x)$$

is called a propositional function,

$$x$$

is its subject, the rest of the property is its predicate

Quantifiers

Once we have a predicate and a domain, we get two (or) new questions,

Definition

The universal quantification of $Q(x)$ is the statement $Q(x)$ is true for **All** x in the domain

We denote this statement by:

$$\forall x Q(x)$$

The above is itself a proposition, and maybe **true** or may be **false**

Definition

The existential quantification of $Q(x)$ is the statement: "There exists (at least 1) x in the domain so that $Q(x)$ is **true**"

We write it as:

$$\exists x Q(x)$$

Informally they correspond to questions

- "Is $Q(x)$ *Always* true?"
- "Is $Q(x)$ *Ever* true?"

Example

$P(x)$ is “ x is wearing red socks”

$Q(x)$ is “ x is wearing socks”

Suppose the domain is all students currently in this room

Observe

Nobody is qualified to answer $\exists xP(x)$ since $\exists xP(x)$ is **false** but nobody can speak for everyone

Similarly one example proves $\exists xQ(x)$ but nobody is qualified to answer $\forall xQ(x)$

Finite Domains

If the domain is finite, say it consists of

$$x_1, x_2, x_3, \dots x_n$$

then

$$\forall xP(x)$$

is a conjunction

$$P(x_1) \wedge P(x_2), \dots \wedge P(x_n)$$

and similarly: $\exists xP(x)$ is a disjunction

$$P(x_1) \vee P(x_2), \dots \vee P(x_n)$$

i.e

$$\sum_{i=1} P(x)$$

Negating Quantifiers

For a finite domain

$$\neg(\exists xP(x)) \equiv \neg(P(x_1) \vee P(x_2) \vee \dots \vee P(x_n))$$

$$\equiv \neg P(x_1) \wedge \neg P(x_2) \wedge \dots \wedge \neg P(x_n)$$

$$\equiv \forall x\neg P(x)$$

We take this as the infinite definition too, so

$$\neg(\exists xP(x))$$

is

$$\forall x(\neg P(x))$$

and similarly

$$\exists x\neg P(x)$$

is a disjunction

$$\neg P(x_1) \vee \neg P(x_2) \vee \neg P(x_3) \vee \dots \vee \neg P(x_n)$$

i.e

$$\sum_{i=1} P(x_n)$$